Water Alarm: A Digital Electroproject

Project Report

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Abstract

This project presents the design and hardware build of a Water Alarm System, a simple sensing-and-alert circuit that warns the user when water reaches a critical level in a tank or accumulates where it should not. A water sensor detects the presence of water and sends a signal to a control circuit built on a PCB. The control circuit then activates an alarm, producing an audible warning through a buzzer and a visual indication through an LED. A relay on the board can additionally switch an external load. The circuit is powered by a separate AC-to-DC power supply unit. The result is a low-cost, easy-to-demonstrate solution that combines sensing, switching, indication and sound output in a single circuit.

1. Introduction

A Water Alarm System is an electronic safety device designed to detect the presence or rising level of water and alert users through an audible alarm. Water leakage and overflowing tanks can cause water wastage, property damage and other problems. Our project uses an electronic control circuit and a power supply to detect water and activate an alarm when water reaches the sensing point.

1.1 Why a water alarm matters

- **Prevent overflow:** Water overflow wastes water and can damage surrounding spaces.
- **Warn before overflow:** An alarm gives an immediate indication before the tank overflows.
- **One complete circuit:** The project demonstrates sensing, switching, indication and sound output in one circuit.

1.2 Key features

- Detects the presence of water.
- Generates an audible warning.
- Uses an electronic control circuit.
- Provides a simple and economical solution.

- Helps prevent water wastage and overflow.

2. Objectives

The primary objective of this project is to develop a simple and effective electronic water detection and warning system. The specific objectives are:

1. To detect the presence of water using a sensing mechanism.
2. To generate an alarm when water is detected.
3. To understand the practical application of electronic components.
4. To design and implement a control circuit using a PCB.
5. To understand the role of a power supply in electronic systems.
6. To develop a low-cost solution for water leakage and overflow warnings.

3. Components Used

No.	Component	Function
1	Control Circuit PCB	Main processing and control unit of the water alarm system
2	Integrated Circuits (ICs)	Perform electronic control and signal-processing functions
3	Relay	Electrically operated switch used to control an external circuit
4	Capacitors	Filtering, smoothing and stabilizing electrical signals and voltage
5	Resistors	Control current and establish appropriate operating conditions
6	LED indicator	Visual indication of the circuit's operating or alarm status
7	Buzzer	Produces an audible warning when the alarm is activated
8	Power Supply Unit	Provides the required electrical power to the control circuit
9	Water Sensor	Detects water and provides a signal to the control circuit

4. Block Diagram and Working Principle

The Water Alarm System operates on the principle of water detection followed by electronic signal processing.

Basic block diagram:

Water Sensor → Control Circuit → Alarm Activation → Buzzer

4.1 Working steps

1. The water sensor is placed at the desired detection point.
2. When water reaches the sensor, it detects the presence of water.
3. The sensor generates an electrical signal.
4. This signal is received by the control circuit.
5. The control circuit activates the alarm mechanism.
6. The buzzer produces sound to alert the user.

The relay can also be used to switch an external electrical load, depending on the circuit connections.

4.2 Background: how water sensing works

A simple water sensor typically uses two exposed conductors (probes). In dry conditions the gap between them is an open circuit. Once water bridges the gap, a small current flows because water containing dissolved salts conducts electricity. The control circuit reads this change and treats it as the "water detected" signal. The slides do not specify the exact probe arrangement, so this section describes the general principle only.

5. Control Circuit PCB

The control circuit PCB is the main part of the project. It contains the electronic components connected together to perform water detection and alarm control.

Main parts visible on the PCB:

- Integrated circuit chips for electronic control (an 8-pin and a 16-pin IC are visible on the board).
- Relay for switching operations.
- Capacitors and resistors for circuit operation.
- LED indicator for visual indication.
- Buzzer connection for audible alerts.
- External sensor connections.

- A voltage regulator and DC input jack for powering the board.

Function: The PCB receives the signal from the water sensor and controls the alarm response. It coordinates the operation of the electronic components to provide an effective warning system.

6. Power Supply Unit

The power supply unit provides the electrical energy required for the operation of the water alarm system. The project uses a separate power supply circuit connected to the control circuit.

Main functions:

1. Receives electrical input from the AC supply.
2. Converts electrical power into a suitable form for the electronic circuit.
3. Provides the required DC output.
4. Helps the control PCB operate properly.

Importance: A stable and appropriate power supply is essential for the reliable operation of electronic components and the overall functioning of the project.

7. Hardware Build

The hardware consists of two parts: the **control circuit PCB**, which holds the ICs, relay, capacitors, resistors, LED and buzzer/sensor connections, and the **power supply**, an enclosed AC-to-DC module with a mains plug whose DC output feeds the control PCB.

8. Applications

- Water tank overflow detection.
- Leakage detection in homes and buildings.
- Monitoring water accumulation in basements.
- Water detection in storage areas.
- Industrial water monitoring systems.

9. Advantages

- Simple electronic design.
- Provides an audible warning.
- Helps reduce water wastage.

- Economical and practical solution.
- Easy to understand and demonstrate.

10. Limitations and Future Scope

(This section is added as general suggestions and is not part of the original slides.)

- The alarm only warns the user; it does not stop the water inflow by itself. The relay could be wired to cut off a pump or valve automatically.
- A single sensing point detects only one level. Multiple probes could indicate low, medium and high levels.
- The mains-powered supply could be complemented with a battery backup so the alarm still works during power cuts.
- Wireless or mobile notifications could alert users who are away from the tank.

11. Conclusion

The Water Alarm System demonstrates the practical application of electronic components in water detection and warning systems. Through this project, our team gained practical knowledge about PCB circuits, integrated circuits, relays, power supply units and sensor-based electronic systems. This project provides a basic and useful approach to detecting water and alerting users, helping prevent water-related problems.